CSCS
Centro Svizzero di Calcolo Scientifico
Swiss National Supercomputing Centre

ETH zürich

Evolving Sarus to augment Podman for HPC

CANOPIE-HPC Workshop @ SC25

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Why Podman in HPC?

- Opening to an ecosystem (community, standards, interfaces)

- Integrations for seamless developer productivity

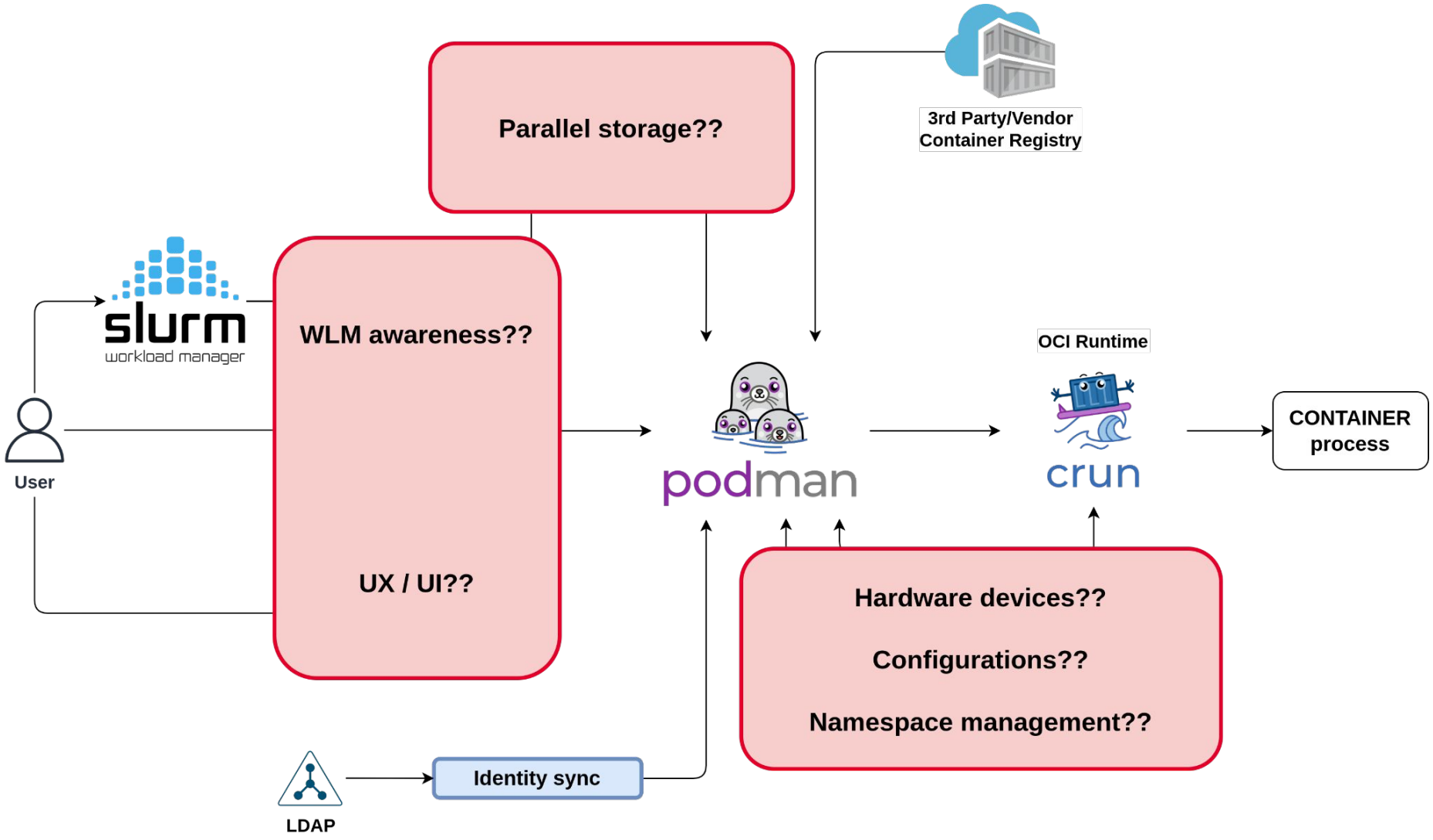
- Technology alignment for HPC, system management and more

- Path towards increased cloud-native adoption

Problems in employing Podman for HPC

- Image storage not supported on parallel filesystems
- Expose devices, hardware, env vars, configs, …
- Namespace management to enable communication (e.g. MPI)
- WLM awareness for efficiency and intra-node cooperation
- All of the above complicate the UX / UI

Complementing Podman with HPC-focused components and know-how



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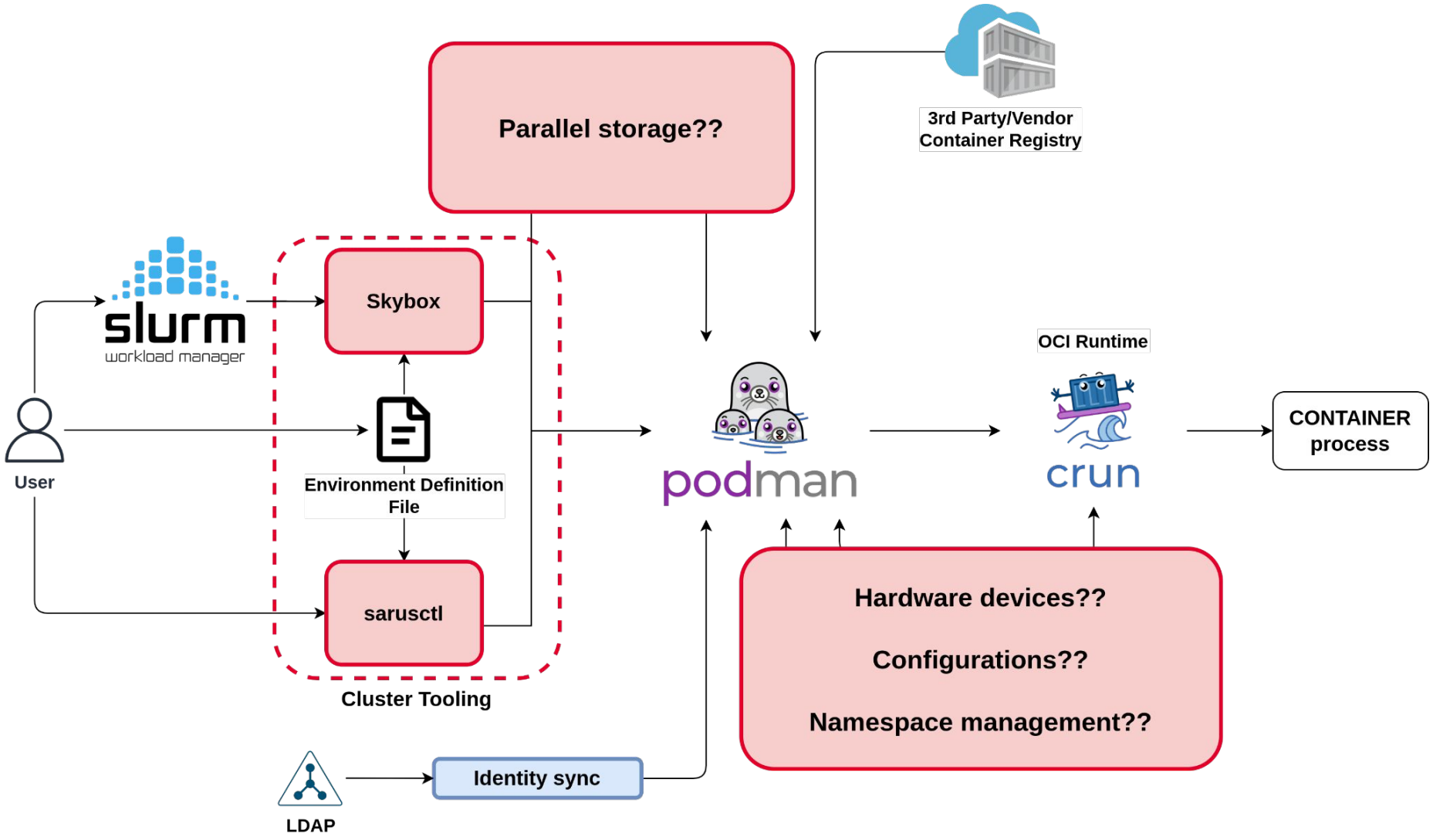

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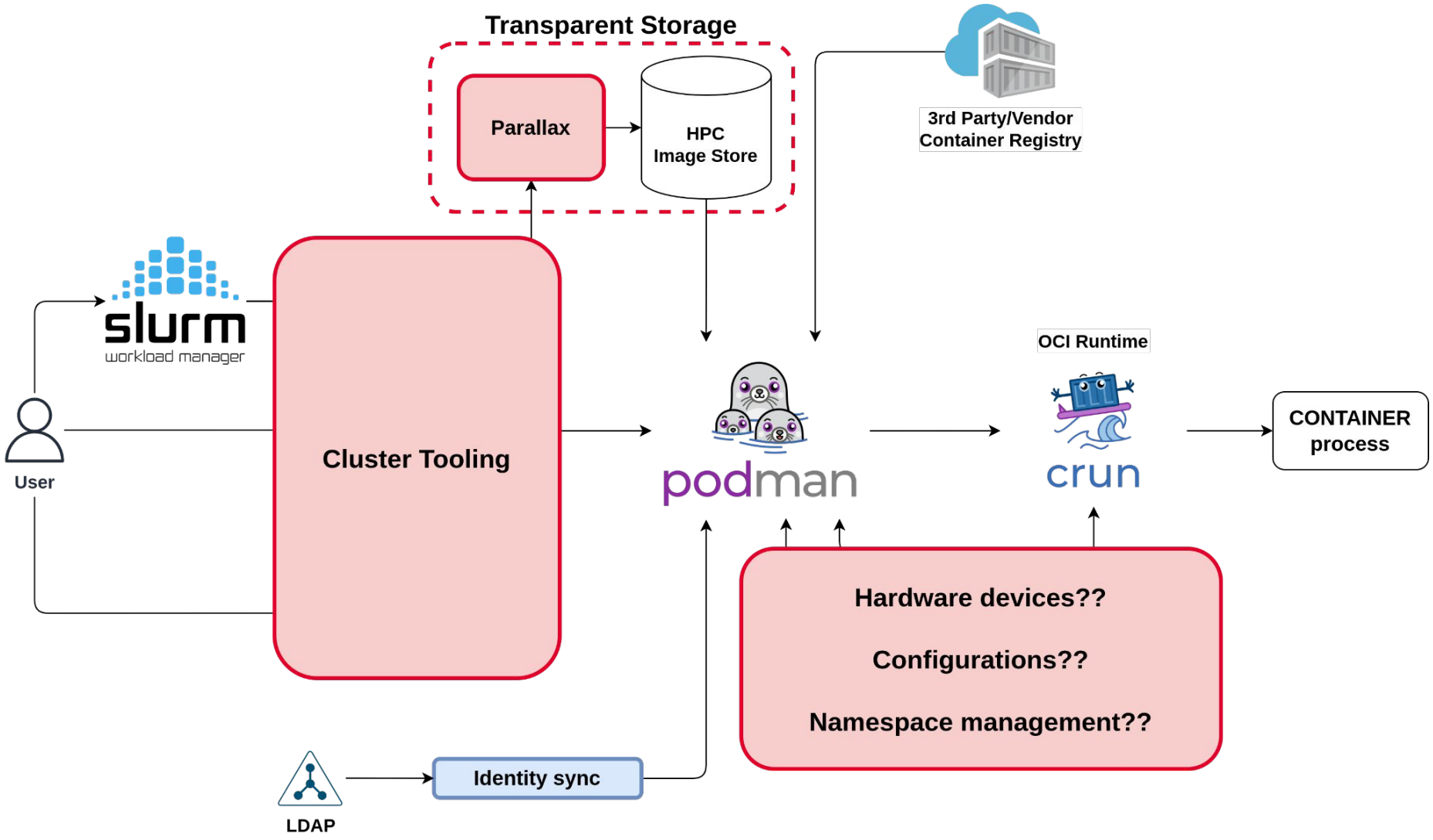

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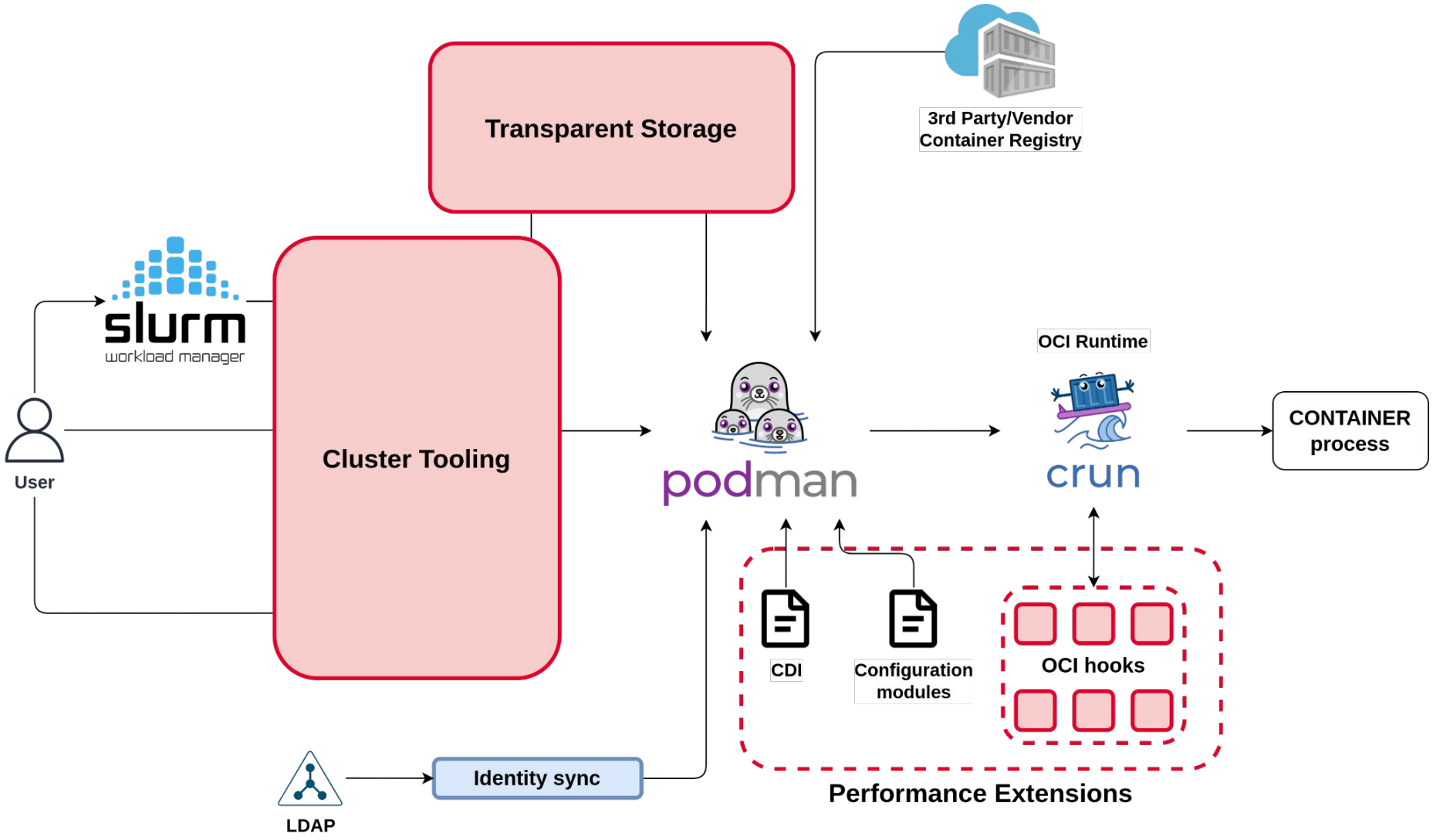

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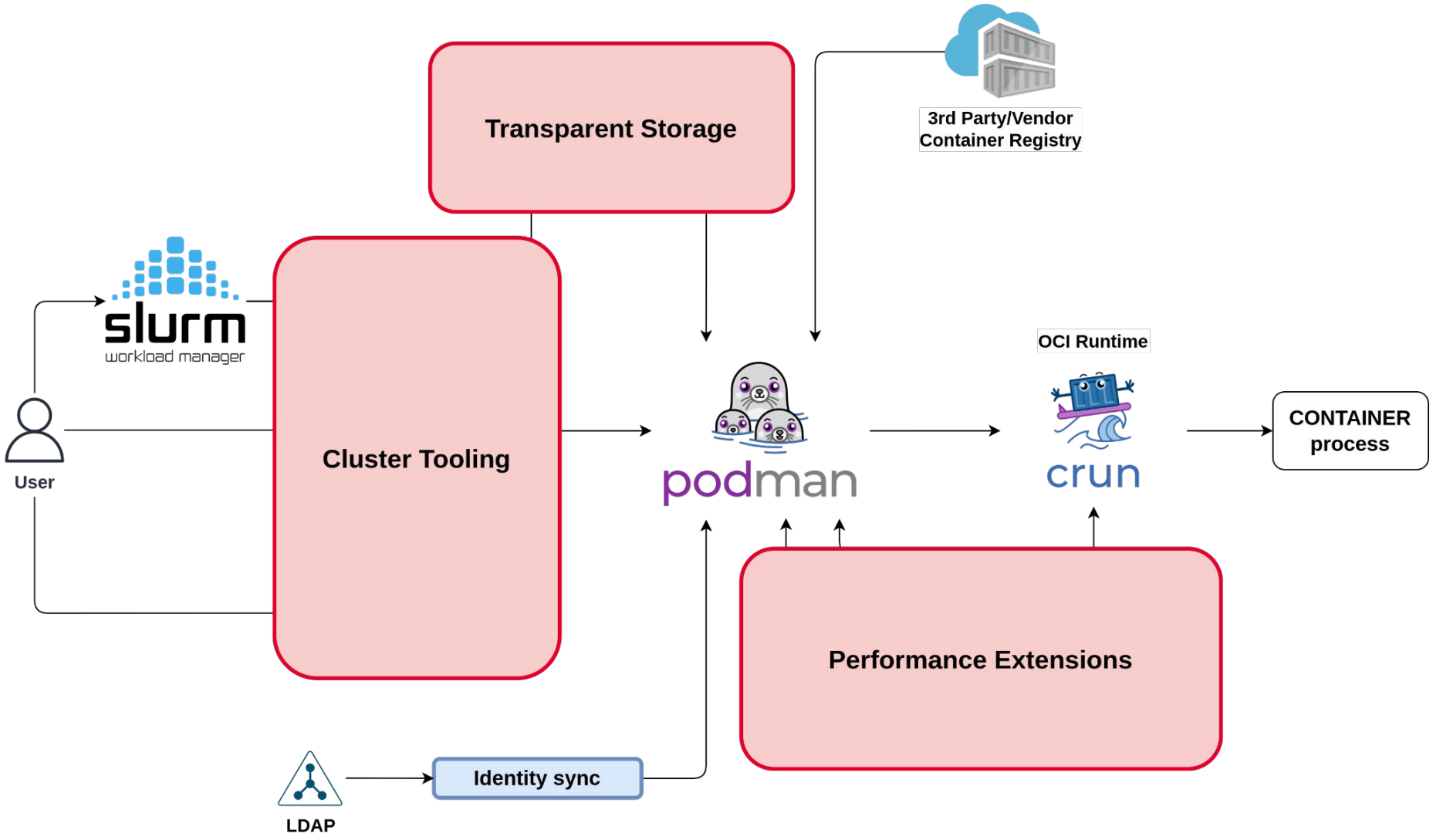
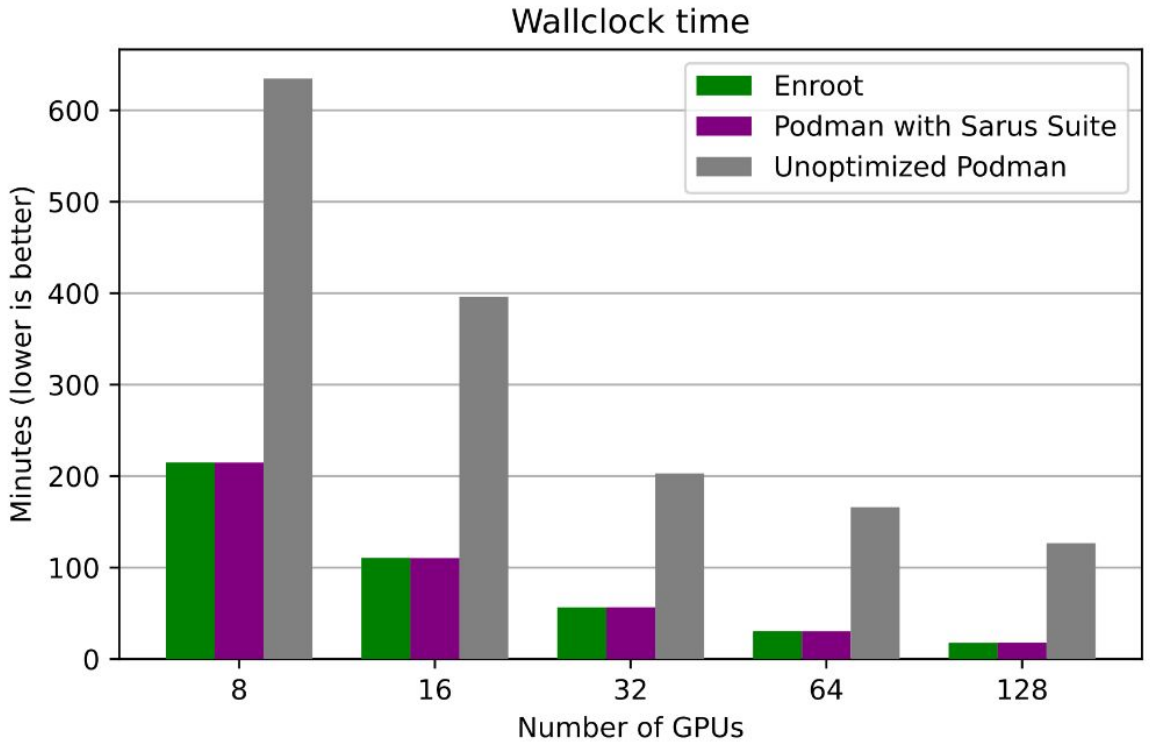


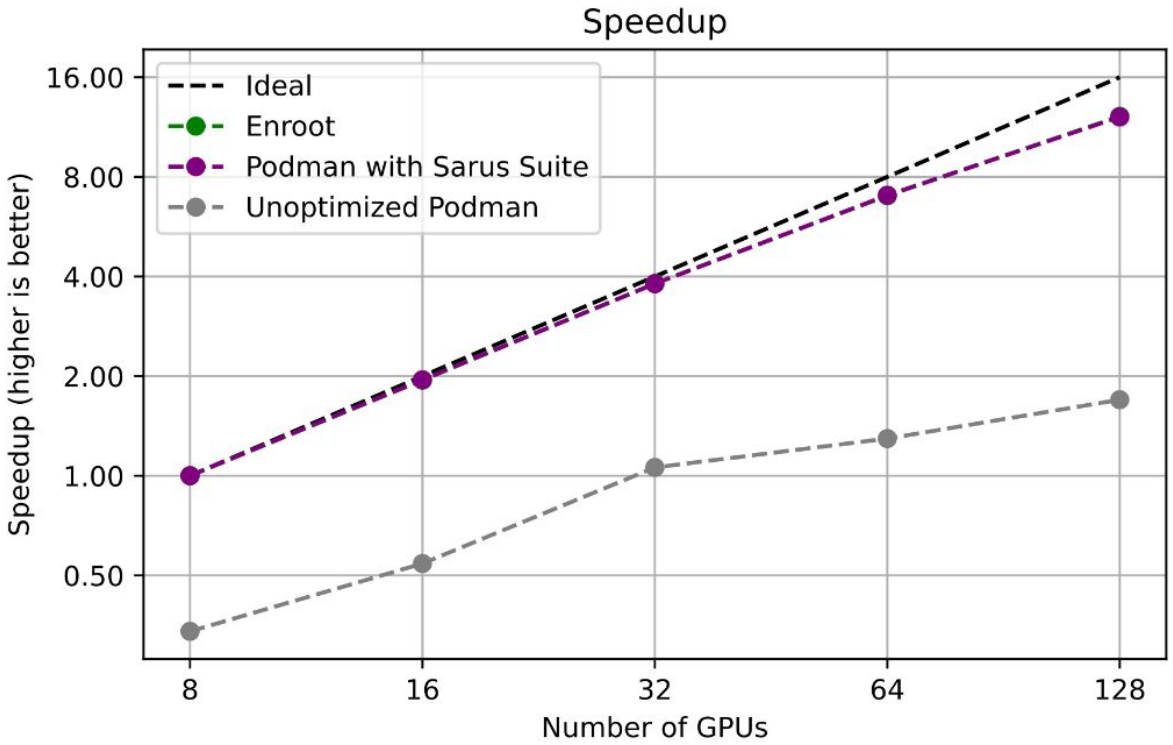
Introducing Sarus Suite

Modular components to unlock the power of HPC for container tools based on open standards.

- **Cluster Tooling**
  - Slurm plugin
  - CLI tool
- **Performance Extensions**
  - OCI hooks, CDIs, Podman config modules
- **Transparent Storage**
  - Parallax storage tool
- **Dev & Admin Tools**
  - Deployment & automation

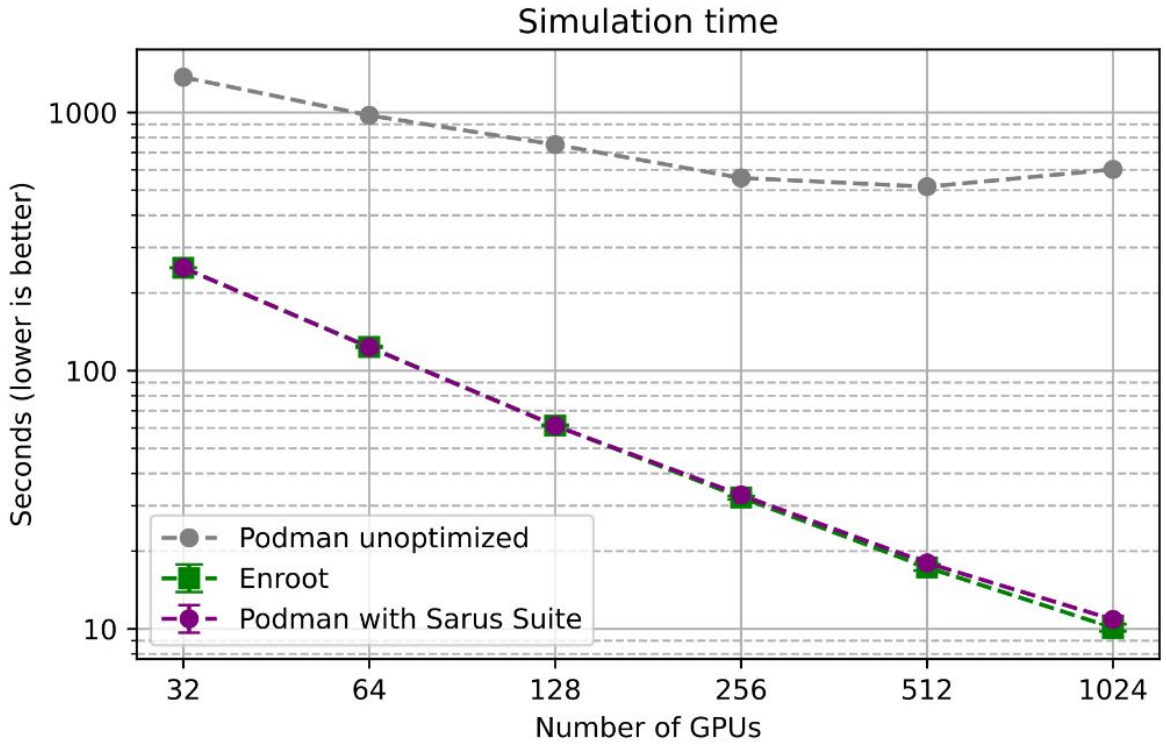
Nanochat LLM pre-training

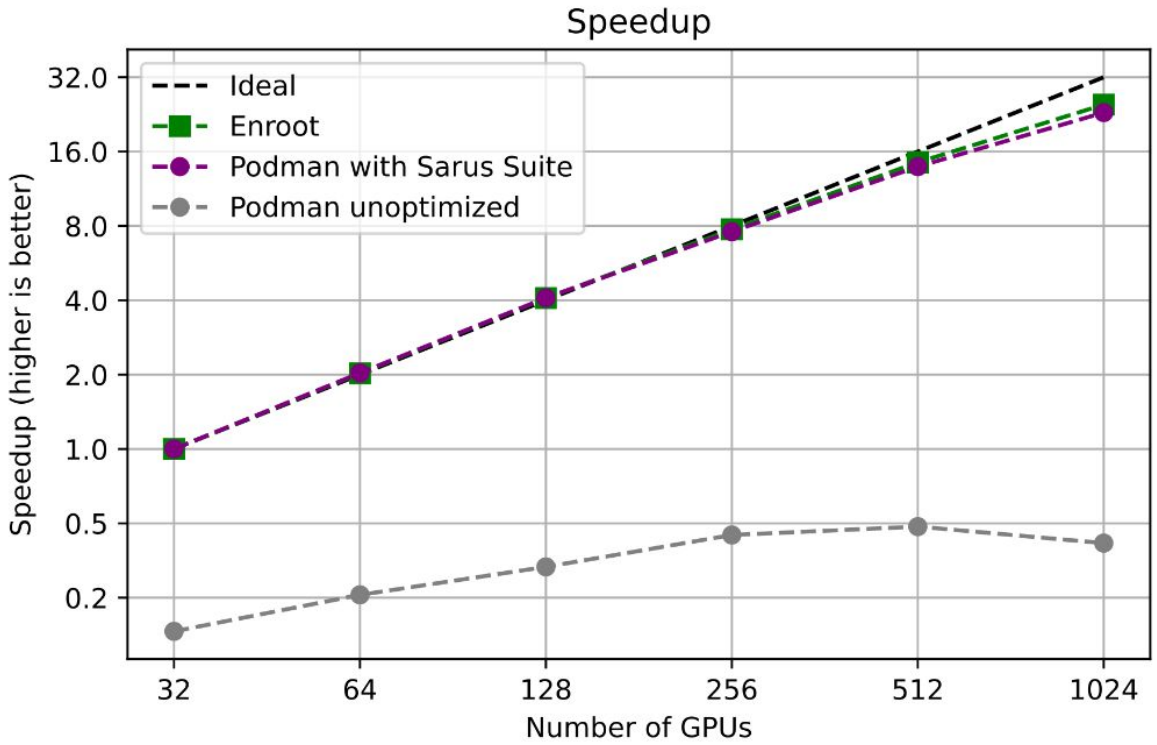




Container: NVIDIA NGC PyTorch 25.10, CUDA 13, Ubuntu 24.04
Test case: Nanochat d20 pre-training (a.k.a. base train)
System: Alps Infrastructure - Machine Learning vCluster (4 x NVIDIA GH200, Slingshot 11 200Gb)

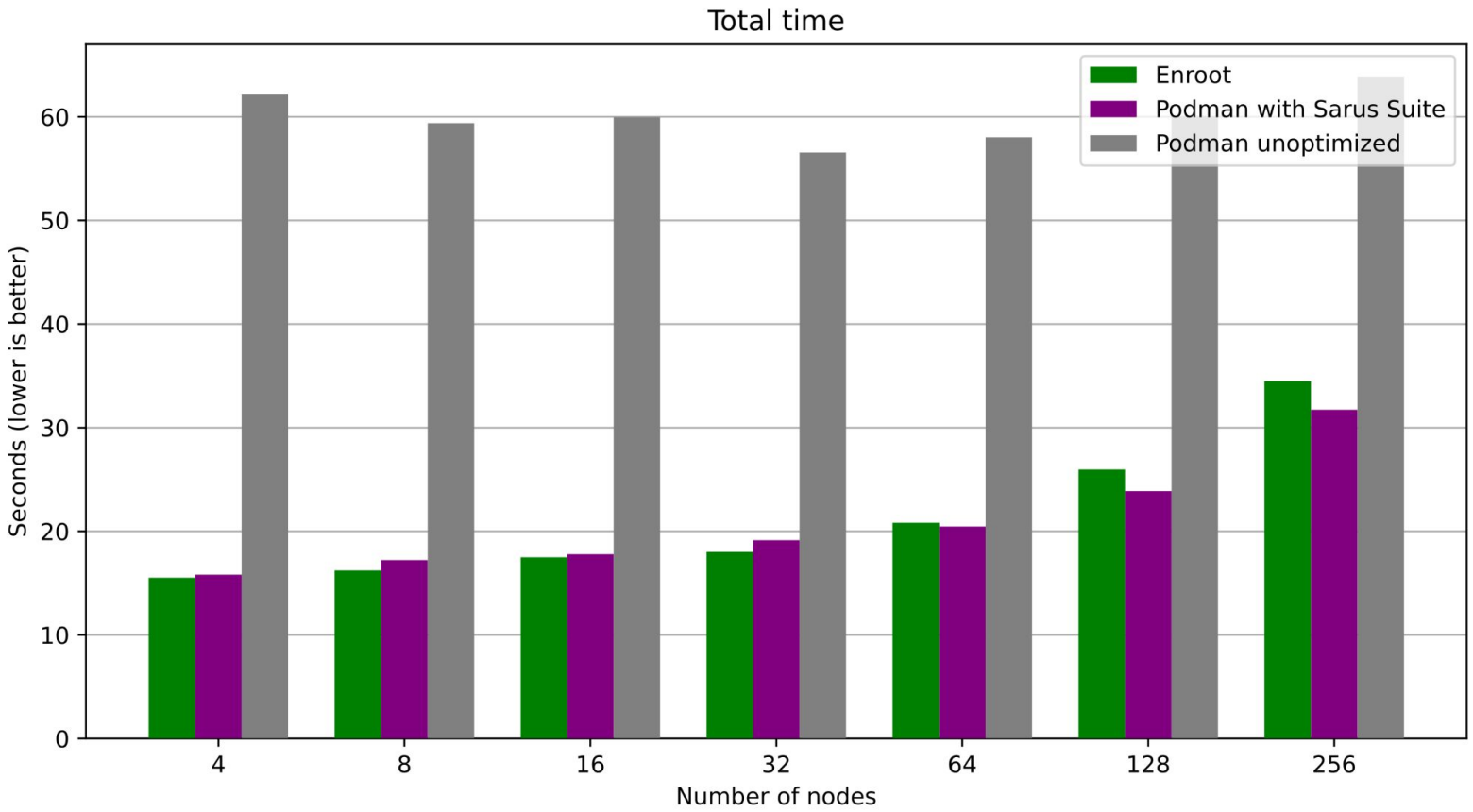
PyFR (Flux Reconstruction CFD)





Container: PyFR 2.1, OpenMPI 5.0.7, CUDA 12.8, Ubuntu 24.04
Test case: 3D Taylor-Green vortex (4 ranks per node, 10 repetitions)
System: Alps Infrastructure - Machine Learning vCluster (4 x NVIDIA GH200, Slingshot 11 200Gb)

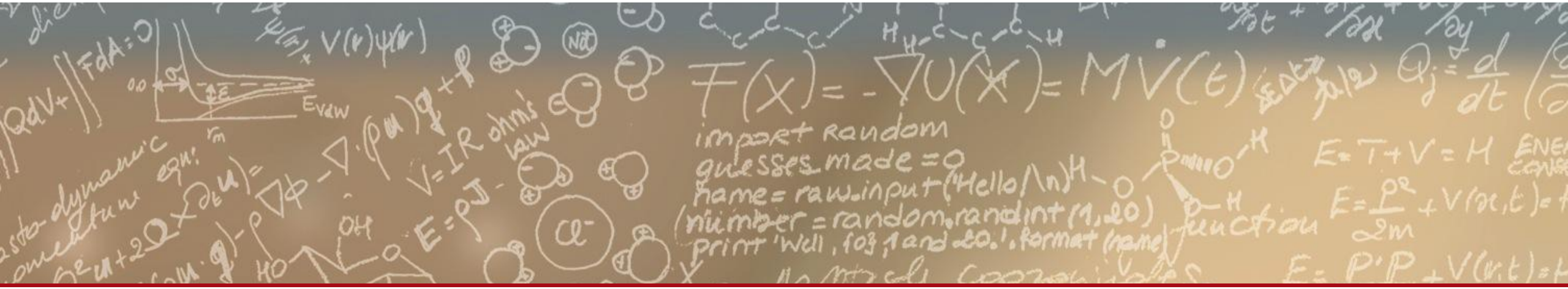
Pynamic (storage filesystem benchmark)



Container: Pynamic w/ Python 3, OpenMPI 5.0.7, Debian 12 (Bookworm)
Test case: Synthetic benchmark simulating import and visit of 495 Python modules with avg. 1850 functions each
System: Alps Infrastructure - Machine Learning vCluster (4 x NVIDIA GH200, Slingshot 11 200Gb)

Conclusions

- Sarus evolving from an engine to a suite of tools
  - Modular, open-source collection of reusable components
  - Continuing to combine HPC with cloud-native flexibility and standards
- Enabling first-class container integration with HPC
  - Containers are transparent, manageable, and fully integrated into HPC
- Strategic shift towards performance-enhanced Podman as runtime
  - CSCS will use Sarus Suite components to enable profitable usage of Podman in HPC
- Forward-looking and OSS-oriented
  - Towards a container stack for adaptability and sustainability
  - Smoother adoption path to other cloud native technologies



Thank you for your attention.